

Lecture 3:

Characteristics of Games and (Skill) Depth

Game Design Spring 2026

Julian Togelius and Tim Merino

(With thanks to Graham Todd and Zehua Jiang)

CHARACTERISTICS OF GAMES

George Skaff Elias, Richard Garfield, and K. Robert Gutschera
foreword by Eric Zimmerman | drawings by Peter Whitley

CHARACTERISTICS OF GAMES

Elias, Garfield, and Gutschera

ADVANCED Game Design

Game Programming Patterns

SWINK
Semiology of Graphics

GAME FEEL

Robert Nystrom

national bestseller



Yannakakis · Togelius



Artificial Intelligence and Games

Second Edition

Yannakakis · Togelius



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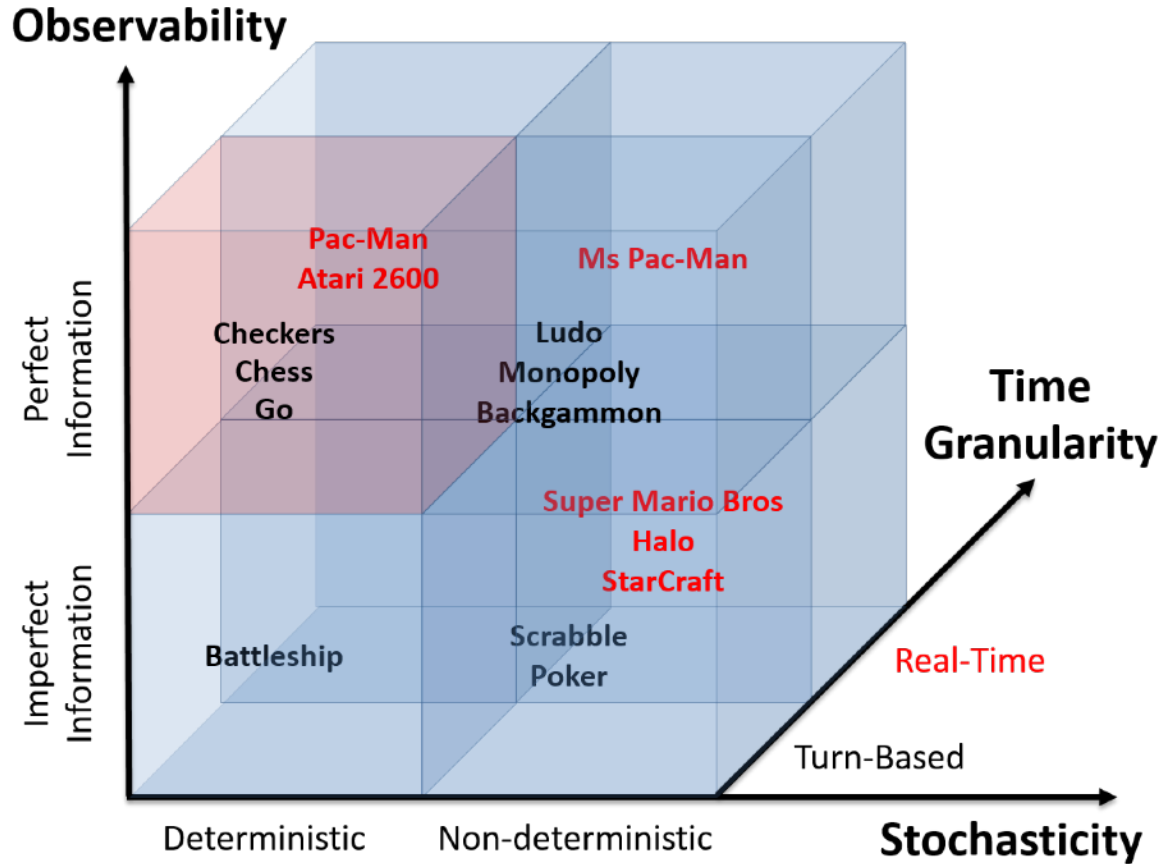
Artificial Intelligence and Games

Second Edition

Georgios N. Yannakakis
Julian Togelius

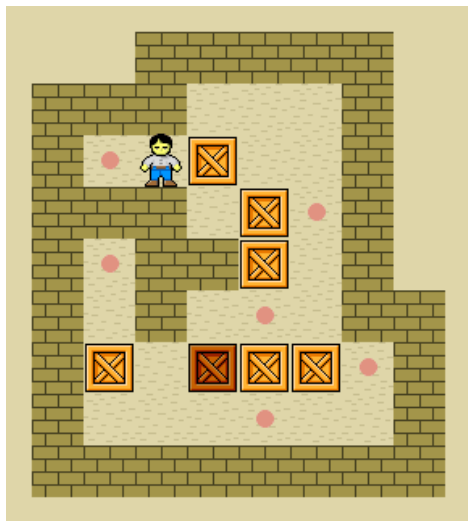
 Springer

- ❑ Stochasticity
vs. Observability
vs. Time Granularity
- ❑ Length of playtime
- ❑ Systems
- ❑ Single vs. Multiplayer
- ❑ Heuristics
- ❑ Dexterity vs. Strategy



Stochasticity vs. Observability vs. Time Granularity

- *stochasticity* - will different outcomes happen from taking an action or the same actions every time?
 - deterministic vs. non-deterministic



Deterministic (Sokoban)



Non-Deterministic (Caveblazers)

Stochasticity vs. Observability vs. Time Granularity

- *observability* - how much of the game is known vs. unknown
 - perfect information vs. imperfect information



Perfect Information (Chess)



Imperfect Information (Battleship)

Stochasticity vs. Observability vs. Time Granularity

- *time granularity* - can you take as much time as you want to make an action or does it play in real-time?
 - turn-based vs. real time



Turn-based (Final Fantasy 3)



Real Time (Elden Ring)

Length of Playtime

- *round* - short and quick relative to the entire gameplay
 - *session* - can vary in time from a few minutes to a few hours
 - “*full game*” - encapsulates the entire gameplay from start to finish
-
- Note: A *session* can have multiple *rounds* and a *full game* can contain many sessions



Length of Playtime



Pokemon



Round
(1 battle)



Session
(Fighting a Gym)



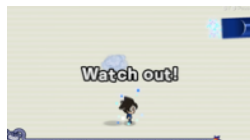
Full Game
(Collecting badges)

Length of Playtime

WarioWare



Round
(1 minigame)



Session
(Multiple minigames)



Full Game
(Playing all minigames)

Systems

- how game elements interact with each other
 - player, object, enemy, ally, abstract items, etc

1. *conditional*

- a. if X, then Y

2. *combination*

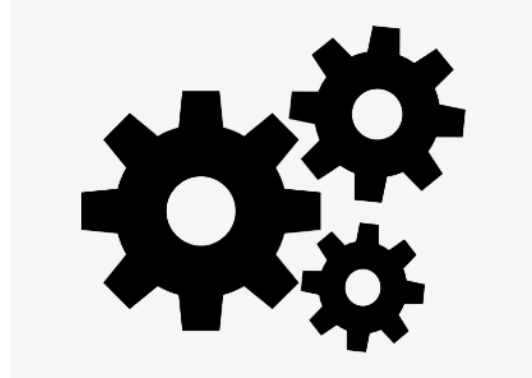
- a. X + Y, makes Z

3. *feedback loops*

- a. AI behavior + state trees; if X, then Y, else Z

4. *resources*

- a. turn # X into ## Y; use X to Y



Systems

Zelda: Breath of the Wild



Conditional: if X, then Y



Feedback Loop: AI behavior + state
trees; if X, then Y, else Z



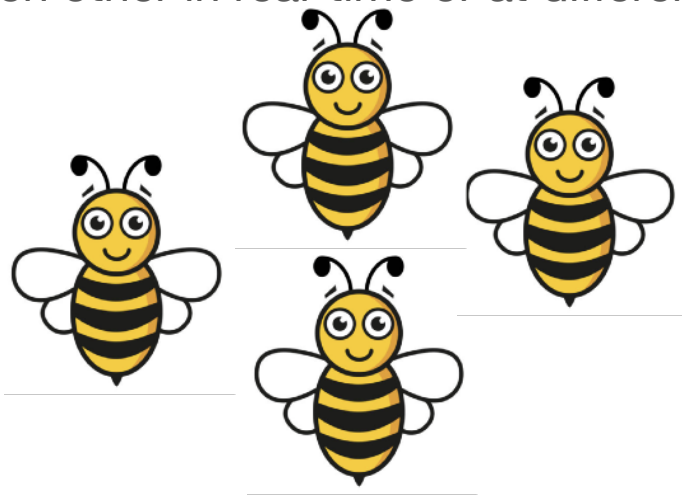
Combination: X + Y, makes Z



Resources: turn # X into ## Y;
use X to Y

Single vs Multiplayer

- *single-player* - can someone play the game by themselves?
 - *one-and-a-half player* - one player plus non-trivial NPCs
- *multi-player* - can multiple people play the game?
 - co-op - players work together towards a goal
 - competitive - players play against each other for the same goal
- *Sync vs. Async* - can players interact with each other in real-time or at different times?



Single vs Multiplayer (in the same game)

Mario Party



Single player



vs



Multiplayer
(competitive)



Single vs Multiplayer (in the same game)

Super Mario Galaxy



Single player



vs

Multiplayer
(co-op)



Single vs Multiplayer (in the same game)

Overcooked



Single player



vs



Multiplayer
(co-op)



Single vs Multiplayer (in the same game)

Minecraft



Single player



Multiplayer
(co-op and competitive)



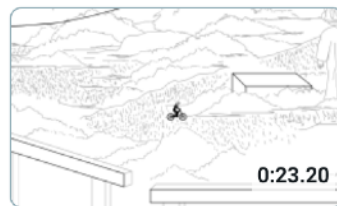
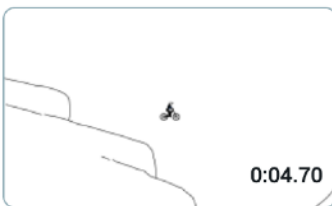
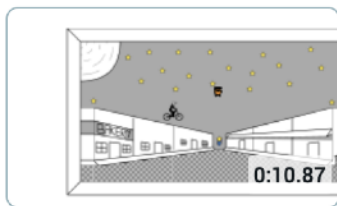
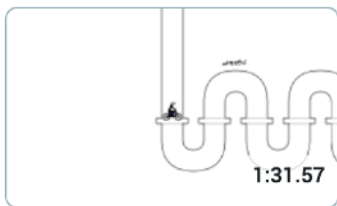
Async Online Multiplayer Mechanics



Messages (Elden Ring)



Design codes (Animal Crossing)



Level editors / creation (Free Rider)

Depth in Strategic Games

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Abstract

This paper explores the question of whether it’s possible to discover a well-defined property of game systems that corresponds to what game designers and players mean by the term “depth.” We propose a measurable property of a game’s formal system, which we call ‘*d*’, that corresponds to the capacity of a game to absorb dedicated problem-solving attention and allow for sustained, long-term learning. To define this property we develop a formal model that measures how susceptible a game is to partial solutions under conditions of steadily increasing computational resources. We then sketch out several directions for using the model to investigate questions about the structural properties of games that produce these effects.

One challenge with the informal use of the word “depth” is that it is often used as a binary term, implying a quality that games either do or don’t have. The real picture is much more complex; every game exists along a spectrum of depth. Moreover, the same game may exhibit different levels of depth in relation to different player communities, or in relation to the same community at different times. We are interested in examining depth as a quality that all games have to various degrees, understanding how this quality is related to a game’s formal structure, and developing conceptual tools that allow us to explore this relationship with greater precision.

Depth is often referred to by game developers (Pulsipher and Others 2011; Kiley 2013; Ghostcrawler 2016) and in scholarly research (Browne 2008; Nielsen et al. 2015;

Heuristics

A **heuristic** is a rule-of-thumb that's generally (but not always) useful.

In games, heuristics help you **narrow down the search over possible actions**. A game without any heuristics can be dull because there's no good way to think through what to do, but a game where all you need is one heuristic might be trivial.

What are some heuristics for Tetris?

What makes a good heuristic?

A good heuristic in games is one that...

1. Applies at every stage of the game (beginning, middle, end)
2. Lies somewhere between gut feeling and brute force
3. Provides some “useful compression” of the game state

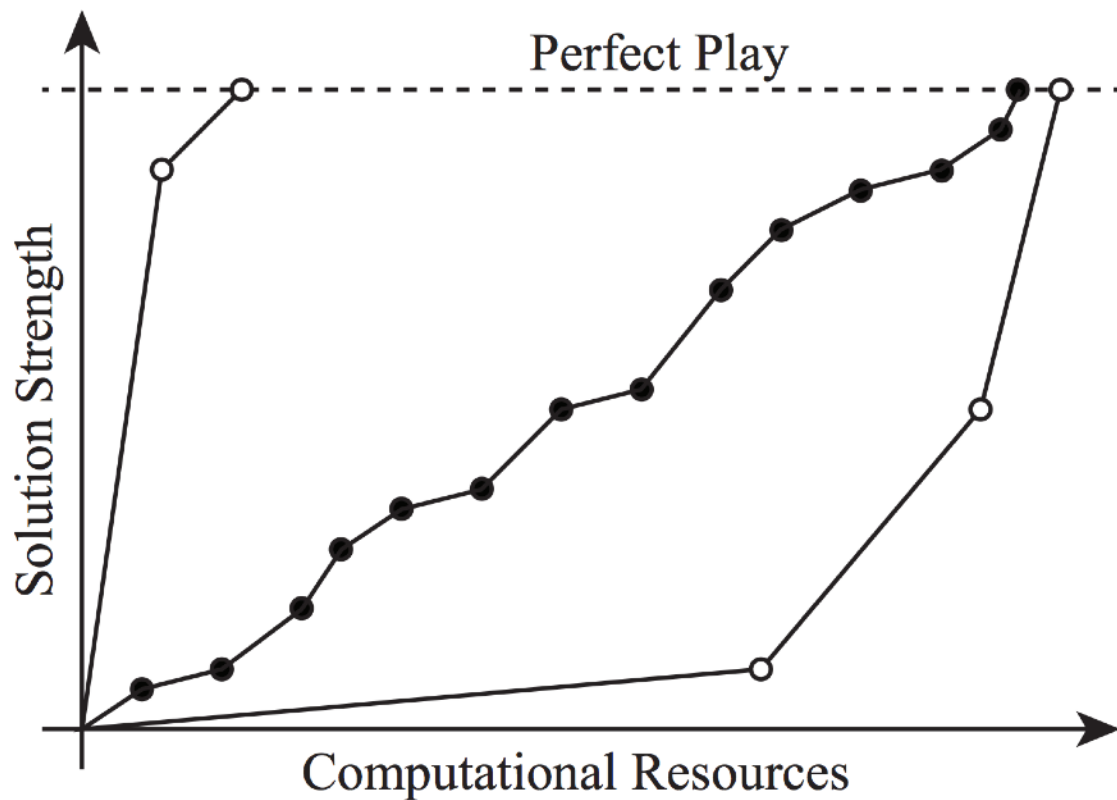


Figure 1: Strategy ladder steps and solution strength, as function of computational resources

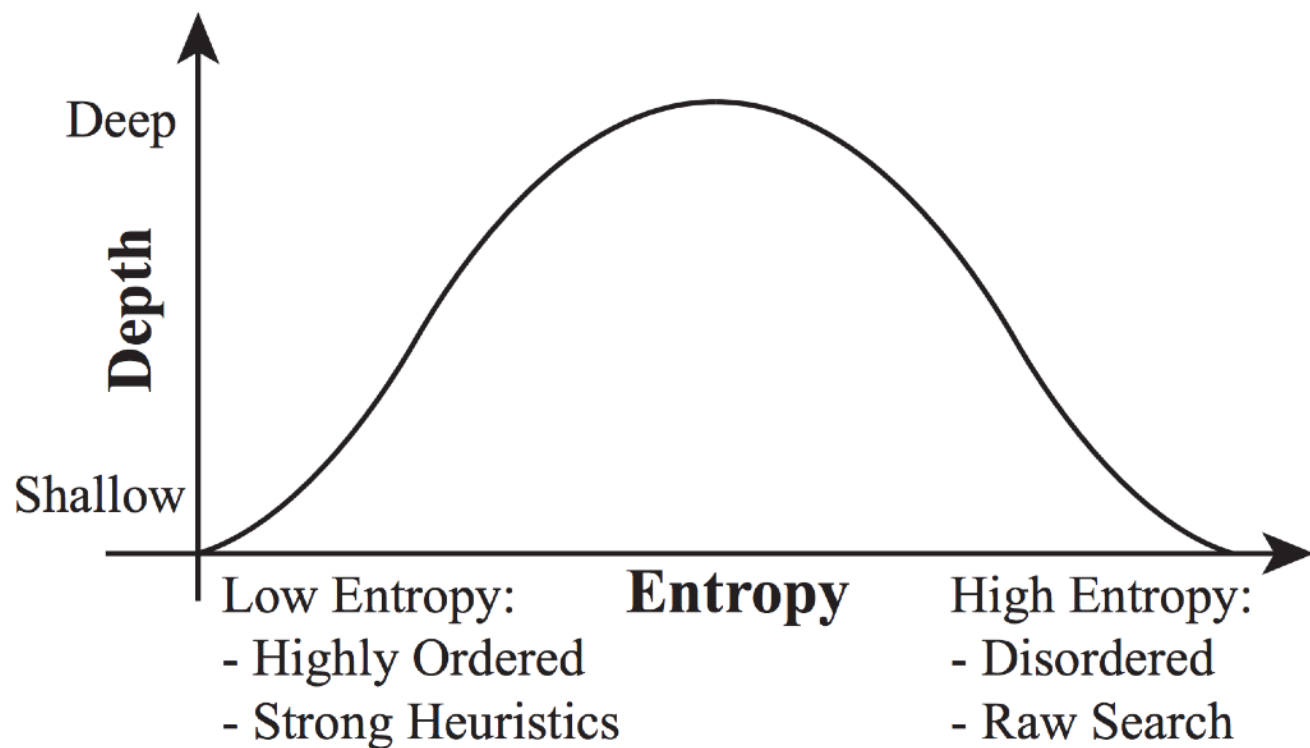
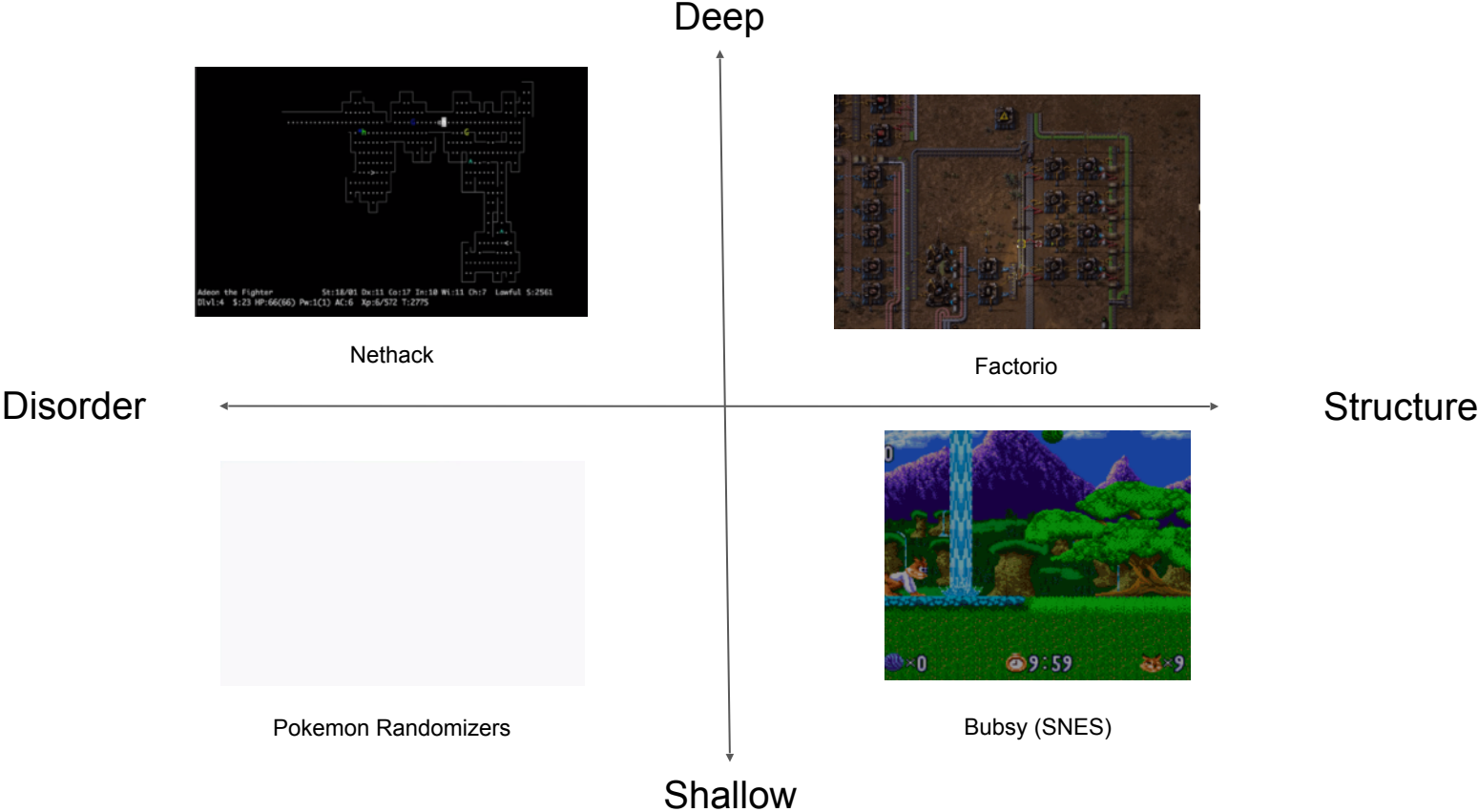


Figure 2: Maximizing depth requires a balance between heuristics and search.

Depth vs. Entropy



Dexterity vs Strategy

dexterity - requires physical / reflexive ability to play well

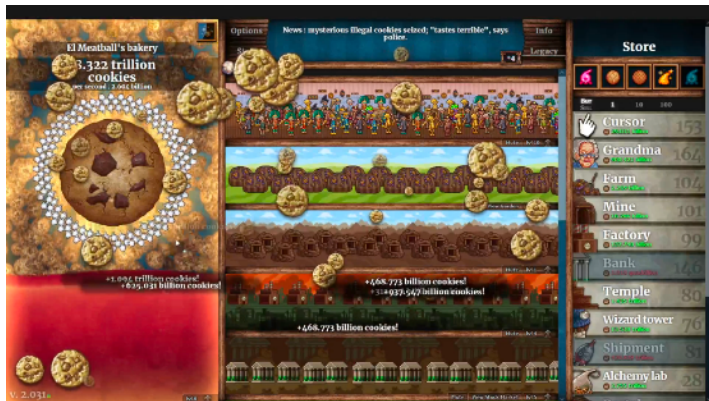
- point and click, button mash, combos, etc.

strategy - requires mental planning ability to play well

- thinking steps ahead, resource management, risk-taking, etc.

Dexterity

Cookie Clicker



Low Dexterity

Elden Ring



High Dexterity

Strategy

Craps

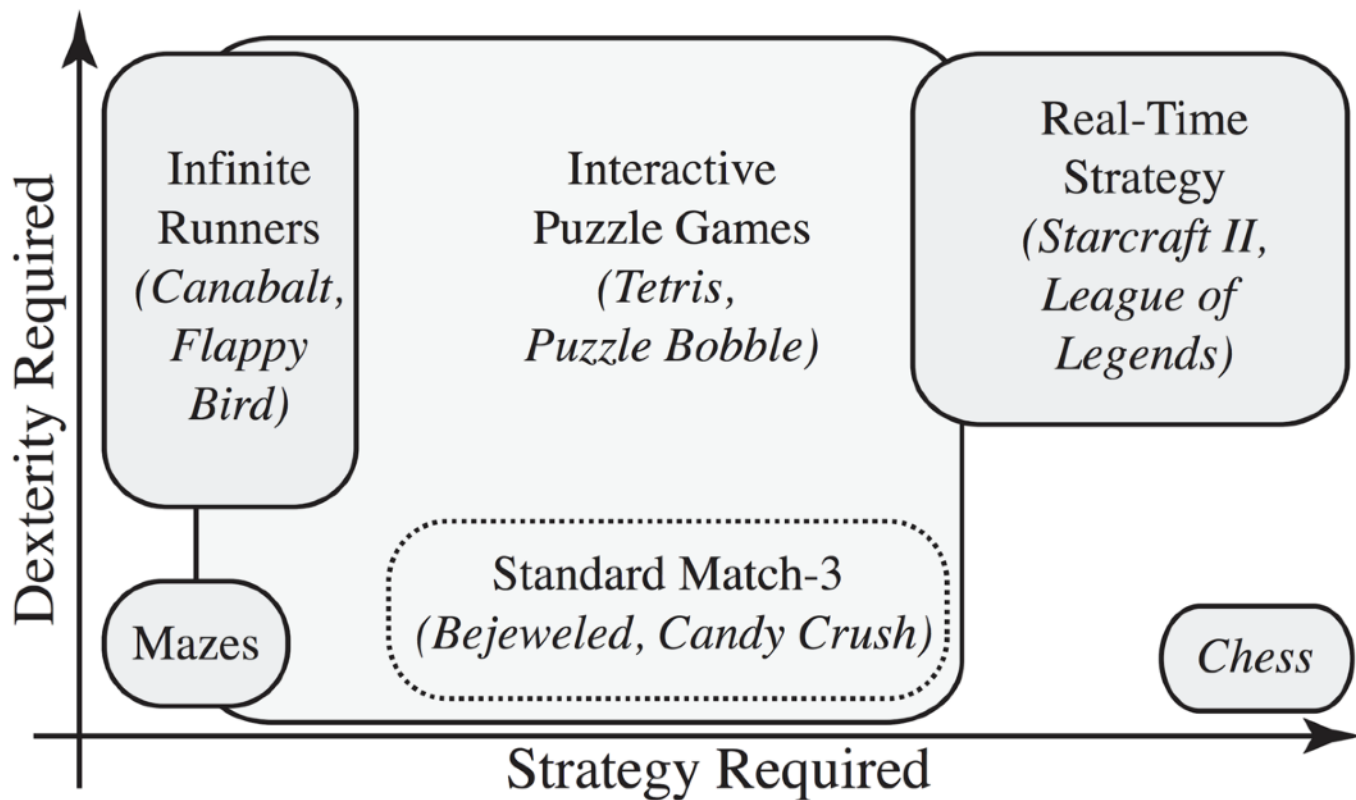


Low(er) Strategy

Starcraft (RTS)



High Strategy



This Week's Prototype

For next class: make a prototype that exhibits **strategic depth**. In other words, the more you think about the game the better you should do. Explain which characteristics you use and which heuristics are learned.

Don't go overboard on mechanics or scope!

This is difficult! Playtests with people outside your group will help.

Use of AI: optionally, yet judiciously. Don't lose control.